

Functions



Outline

The concept of function

Definition of function, domain, codomain and image

Surjections, injections and bijections

The concept of function



Introduction: The concept of function

- ▶ In this section we introduce the concept of **function** in an intuitive way
- ▶ We start from concrete examples (a falling body, an investment) in which one quantity **depends** on another
- ▶ We will understand that what characterizes a function is the **uniqueness** of the relation between input and output

Variable quantities

The existence of a **variable quantity** implies the existence of a **relation** between **two quantities**, that is, the dependence of one quantity on another.

- ▶ In each case, this relation follows a certain **law** or **formula**.

Example (1/2): laws/formulas $\rightarrow$ functions

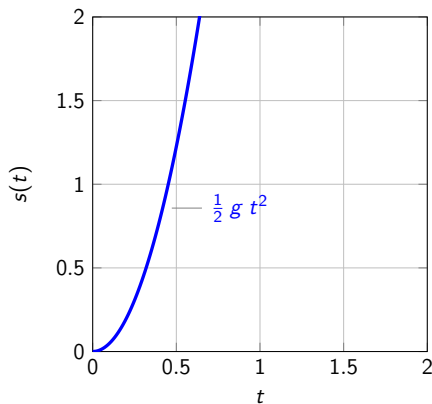
If a heavy object is dropped from a certain height, the distance traveled by the object varies with time t according to the formula:

$$s(t) = \frac{1}{2} g t^2 \quad \text{with } t \geq 0$$

where $g \approx 9,8$ is the gravitational acceleration constant.

Hence, the time t is associated with the distance traveled $s(t)$:
 $t \mapsto s(t)$.

Example (2/2): laws/formulas $\rightarrow$ functions



Example (1/3): laws/formulas $\rightarrow$ functions

If a unit capital is invested for one year at the annual rate i , with interest paid monthly at a rate $\frac{i}{12}$, we have:

- ▶ after the first month, a capital equal to:

$$1 + \frac{i}{12} \cdot 1 = \underbrace{1 + \frac{i}{12}}_{\alpha}$$

- ▶ after the second month, a capital equal to:

$$\underbrace{1 + \frac{i}{12}}_{\alpha} + \frac{i}{12} \underbrace{\left(1 + \frac{i}{12}\right)}_{\alpha} = \left(1 + \frac{i}{12}\right) \left(1 + \frac{i}{12}\right) = \underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta}$$

Example (2/3): laws/formulas $\rightarrow$ functions

- after the third month, a capital equal to:

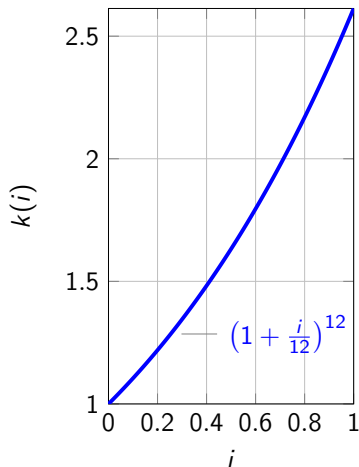
$$\underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta} + \frac{i}{12} \underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta} = \left(1 + \frac{i}{12}\right)^2 \left(1 + \frac{i}{12}\right) \\ = \left(1 + \frac{i}{12}\right)^3$$

Repeating the argument, the capital at the end of the year therefore depends on the rate i according to the formula:

$$k(i) = \left(1 + \frac{i}{12}\right)^{12} \quad \text{with } i \in [0, 1]$$

Hence, the rate i is associated with the capital $k(i)$ at the end of the year: $i \mapsto k(i)$.

Example (3/3): laws/formulas $\rightarrow$ functions



Inputs and outputs

- ▶ In each of the previous examples, a *real number* (**input**) is associated ***uniquely*** with another *real number* (**output**). It is precisely the **uniqueness of the relation** that characterizes a function.
- ▶ In general, the **admissible inputs** for a given relation (function) are subject to natural restrictions, related to the very nature of the relation.

Example: admissible inputs

- ▶ In the first example the “starting” real number has the meaning of *time*; imagining that the object is dropped at an initial time $t = 0$, it is clear that we must restrict ourselves to times $t \geq 0$.
- ▶ In the second example, clearly we must have $0 \leq i \leq 1$, where 1 corresponds to a rate of 100%.

The concept of function:

- ▶ A variable quantity expresses the **dependence** of one quantity on another, according to a **law** or **formula**
- ▶ Examples: the distance $s(t) = \frac{1}{2}gt^2$ traveled by a falling body, the capital $k(i) = \left(1 + \frac{i}{12}\right)^{12}$
- ▶ What characterizes a function is **uniqueness**: each input corresponds to exactly one output
- ▶ The **admissible inputs** are subject to natural restrictions

Definition of function, domain, codomain and image

Introduction: Definition of function, domain, codomain and image

- ▶ In this section we give the **formal definition** of a function between any two sets
- ▶ We introduce the concepts of **domain**, **codomain** and **image**, and the notation $f : A \rightarrow B, x \mapsto f(x)$
- ▶ This language will be used throughout the rest of the course

Definition 1: of domain

*The set of admissible inputs for a given function is called the **domain**.*

- ▶ The expressions **independent variable** (for a generic *input*) and **dependent variable** (for the *output*) are often used.

Example: other types of relations

Let us consider the set of students of a given course.

- ▶ If A is the set of students and B is the set of their names, the relation that associates each student with his or her name is *unique* (single-valued).
- ▶ In this case neither A nor B is a set of numbers, but nevertheless a unique (single-valued) relation between these two sets is well defined.

On the other hand, the converse is not necessarily true: there could be two students with the same name.

Definition 2: of function

*Given any two sets A , B , a **function** f with **domain** A and values in B (or also “with **codomain** B ”) is any law that associates with each element of A one and only one element of B .*

Notation

- ▶ The notation:

$$f : A \rightarrow B$$

(which reads “ f defined from A to B ”), indicates the **domain** and the **codomain** of the function f .

- ▶ The notation

$$f : x \mapsto f(x)$$

(which reads “ f maps x to $f(x)$ ”) indicates how the function f acts on the elements.

- ▶ The **symbol** $f(x)$ **denotes the output or the value** that the function f associates with x , and it must not be confused with the symbol f , which denotes the **function itself**.

Example: notation for functions

For the function

$$k(i) = \left(1 + \frac{i}{12}\right)^{12}$$

of the previous example we would have:

$$k : [0, 1] \rightarrow \mathbb{R}$$

$$k : i \mapsto \left(1 + \frac{i}{12}\right)^{12}$$

The notation $y = f(x)$

- The notation

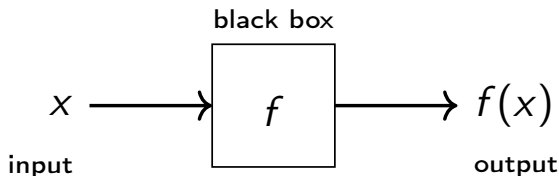
$$y = f(x)$$

is also used to indicate that the variable y is a function of x .
In this case y is the output or the value that f associates with the input x .

For brevity, sometimes the value of $f(x)$ is defined directly in terms of x , as for example $f(x) = x^2 + 10$.

The function as a black box

- In general, we can think of a function as a **black box** that associates with each admissible input x (**input**) a unique output $f(x)$ (**output**):



Definition 3: of image and image of the domain

*The output corresponding to x is called the **image** of x ; the set of possible outputs is called the **image of the domain A under f** and is denoted by the symbol $f(A)$ or $\text{Im } f$.*

- ▶ Note that in the notation $f : A \rightarrow B$, the codomain B can be larger than the image $f(A)$, that is, in general we have

$$f(A) \subseteq B.$$

- ▶ If f is real-valued, we usually write $f : A \rightarrow \mathbb{R}$ without specifying the actual image of f .

Summary

Definition of function:

- ▶ The **domain** is the set of admissible inputs
- ▶ A **function** $f : A \rightarrow B$ associates with each element of A *one and only one* element of B (**codomain**)
- ▶ $f(x)$ is the value associated with x , f is the function itself; we also write $y = f(x)$

Image:

- ▶ $f(A) = \text{Im } f$ is the set of possible outputs and in general $f(A) \subseteq B$

Surjections, injections and bijections



Introduction: Surjections, injections and bijections

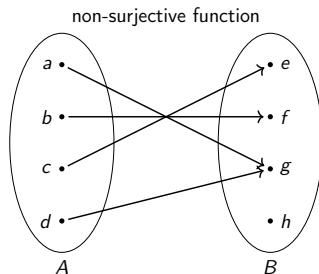
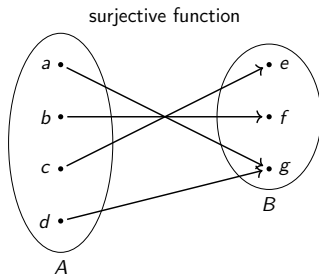
- ▶ In this section we classify functions according to how they relate the domain and the codomain
- ▶ We introduce **surjective**, **injective** and **bijective** functions, with their set representation
- ▶ We prove by induction the relations between the cardinalities of domain and codomain: these concepts will be fundamental for inverse functions

Definition 4: of surjection (surjective function)

*A function f is a **surjection** if the image of its domain coincides with its codomain.*

- ▶ A surjection $f : A \rightarrow B$ is sometimes also called a mapping *onto* B (from A).

Surjection: set representation



Example: surjective and non-surjective functions

- ▶ The function $f(n) = \lfloor \frac{n}{2} \rfloor$ is a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since every element of the codomain $\mathbb{N}$ is the image of some value of the domain.
- ▶ The function $f(n) = 2n$ is not a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since no argument of f produces 3 as a value.
- ▶ The function $f(n) = 2n$ is, however, a surjective function from the natural numbers to the even numbers.

Observation 1

Given two sets A and B , and a function $f : A \rightarrow B$, if f is surjective then $|A| \geq |B|$.

Proof (1/2)

We proceed by induction on the number n of elements in the codomain of the function.

► Base case

If there is only one element in the codomain ($|B| = 1$), since the function is surjective, we have $|A| \geq 1$ (every element of B is the image of at least one element of A). Hence $1 \geq 1$, which is clearly true.

Proof (2/2)

► Inductive step

Assume then that all functions with codomain of size n satisfy $|A_n| \geq |B_n|$.

Considering codomains of size $n + 1$ we have

$|B_{n+1}| = |B_n| + 1$. Since the codomain has one more element and the function is surjective, we have $|A_{n+1}| \geq |A_n| + 1$.

Substituting we obtain:

$$\underbrace{|A_n|}_{\leq |A_{n+1}| - 1} \geq \underbrace{|B_n|}_{= |B_{n+1}| - 1} \quad \text{hence} \quad |A_{n+1}| \geq |B_{n+1}|.$$

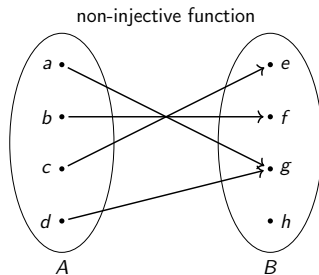
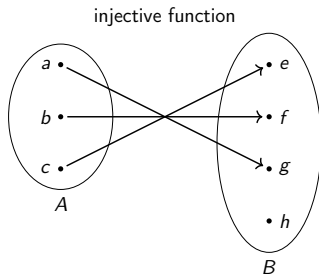


Injection

Definition 5: of injection (injective function)

*A function f is an **injection** if distinct arguments of f produce distinct values, that is, if $a \neq b$ implies $f(a) \neq f(b)$.*

Injection: set representation



Example

- ▶ The function $f(n) = 2n$ is an injective function from $\mathbb{N}$ to $\mathbb{N}$, since each even number b is the image under f of exactly one element of the domain, namely $n = \frac{b}{2}$.
- ▶ The function $f(n) = \lfloor \frac{n}{2} \rfloor$ is not an injective function, since the value 1 is obtained from two arguments: 2 and 3.

Injectons and cardinality

- An injection is also called a **one-to-one** function.

Observation 2

Given two sets A and B , and a function $f : A \rightarrow B$, if f is injective then $|A| \leq |B|$.

Proof (1/2)

We proceed by induction on the number n of elements in the domain of the function.

► Base case

If there is only one element in the domain ($|A| = 1$), since each element of the domain is associated with one and only one element of the codomain, we have $|B| \geq 1$. Hence $1 \leq 1$, which is clearly true.

Proof (2/2)

► Inductive step

Assume then that all functions with domain of size n satisfy $|A_n| \leq |B_n|$.

Considering domains of size $n + 1$ we have $|A_{n+1}| = |A_n| + 1$ and, since the function is injective, $|B_{n+1}| \geq |B_n| + 1$.

Substituting we obtain

$$\underbrace{|A_n|}_{= |A_{n+1}| - 1} \leq \underbrace{|B_n|}_{= |B_{n+1}| - 1} \quad \text{hence} \quad |A_{n+1}| \leq |B_{n+1}|.$$

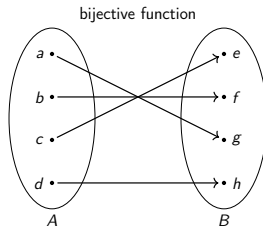


Bijection

Definition 6: of bijection (bijective function or one-to-one correspondence)

A function f is a **bijection** if: (i) it is injective and (ii) it is surjective.

► Set representation:



Example

- ▶ The function $f(n) = (-1)^n \lceil \frac{n}{2} \rceil$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}$.
The values of the function are:

$$f(0) = 0, \quad f(1) = -1, \quad f(2) = 1, \quad f(3) = -2, \quad f(4) = 2 \dots$$

- ▶ The function is injective, since no element of $\mathbb{Z}$ is the image of more than one element of $\mathbb{N}$.
- ▶ The function is surjective, since all elements of $\mathbb{Z}$ are images of some element of $\mathbb{N}$.

Bijections and permutations

- ▶ A bijection is also called a **one-to-one** correspondence, since it pairs elements of the domain with elements of the codomain.
- ▶ A bijection from a set A to itself is also called a **permutation**.

Surjections, injections and bijections:

- ▶ **Surjective:** the image coincides with the codomain; if f is surjective then $|A| \geq |B|$
- ▶ **Injective:** $a \neq b \Rightarrow f(a) \neq f(b)$; if f is injective then $|A| \leq |B|$
- ▶ **Bijjective:** injective and surjective; a bijection from A to itself is a **permutation**
- ▶ Example: $f(n) = (-1)^n \lceil \frac{n}{2} \rceil$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}$

Final summary

- ▶ A **function** associates with each admissible input *one and only one* output
- ▶ $f : A \rightarrow B$: A is the **domain**, B the **codomain**, $f(A) \subseteq B$ the **image**
- ▶ Functions can be **surjective**, **injective** or **bijjective**
- ▶ For finite sets: surjective $\Rightarrow |A| \geq |B|$, injective $\Rightarrow |A| \leq |B|$
(proofs by induction)